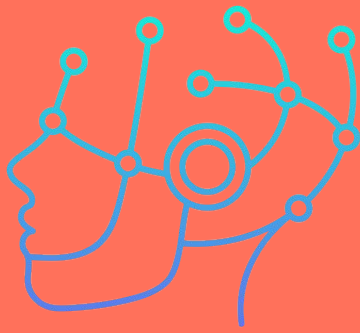




ZAS ROBOTICS
BUILD THE FUTURE

Motion Creativity Board Documentation (v1.0)

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1. Motion Creative kit – Product Overview & Hardware Documentation.

The ZAS Robotics Motion Creative Kit is a complete motion-learning and robotics development platform designed for beginners, students, hobbyists, and engineering educators.

It enables hands-on exploration of DC motors, servo motors, stepper motors, GPS, encoders, I²C sensors, OLED displays, and analog/digital electronics, using a simple plug-and-play interface.

The Motion Creative Board exposes all essential hardware blocks on the front side, while the Arduino UNO is safely mounted at the back side of the board.

This reverse-shield design keeps wiring clean, protects hardware, and ensures a smooth learning experience — ideal for classrooms, labs, workshops, STEM clubs, and self-learning students.

The Motion Creative Kit-3 contains 22 clearly labelled hardware blocks, each responsible for a specific function in robotics and motion control.

Hardware Blocks Overview (22 Functional Sections) The ZAS Robotics Motion Creative Board's features are grouped into 22 blocks.

1. Power Block
2. Arduino Pins Block
3. Basic Electronics Components
4. Potentiometer
5. Free Connectors Block
6. Servo Motor
7. GPS Sensor
8. DC/Encoder Motor
9. Rotary Encoder
10. Encoder/DC Motor Connector
11. TB6612 Motor Driver (DC/Encoder Motor Driver)

12. OLED
13. ON/OFF Switch
14. Reset Button
15. Accelerometer
16. Gesture / I2C Connector
17. Sensor Connector (Signal Pin 12)
18. Mini Breadboard
19. Stepper Motor Driver
20. Servo Connector
21. Stepper Motor Connector
22. Stepper Motor

The functionality and features of each block is discussed in the following sections.

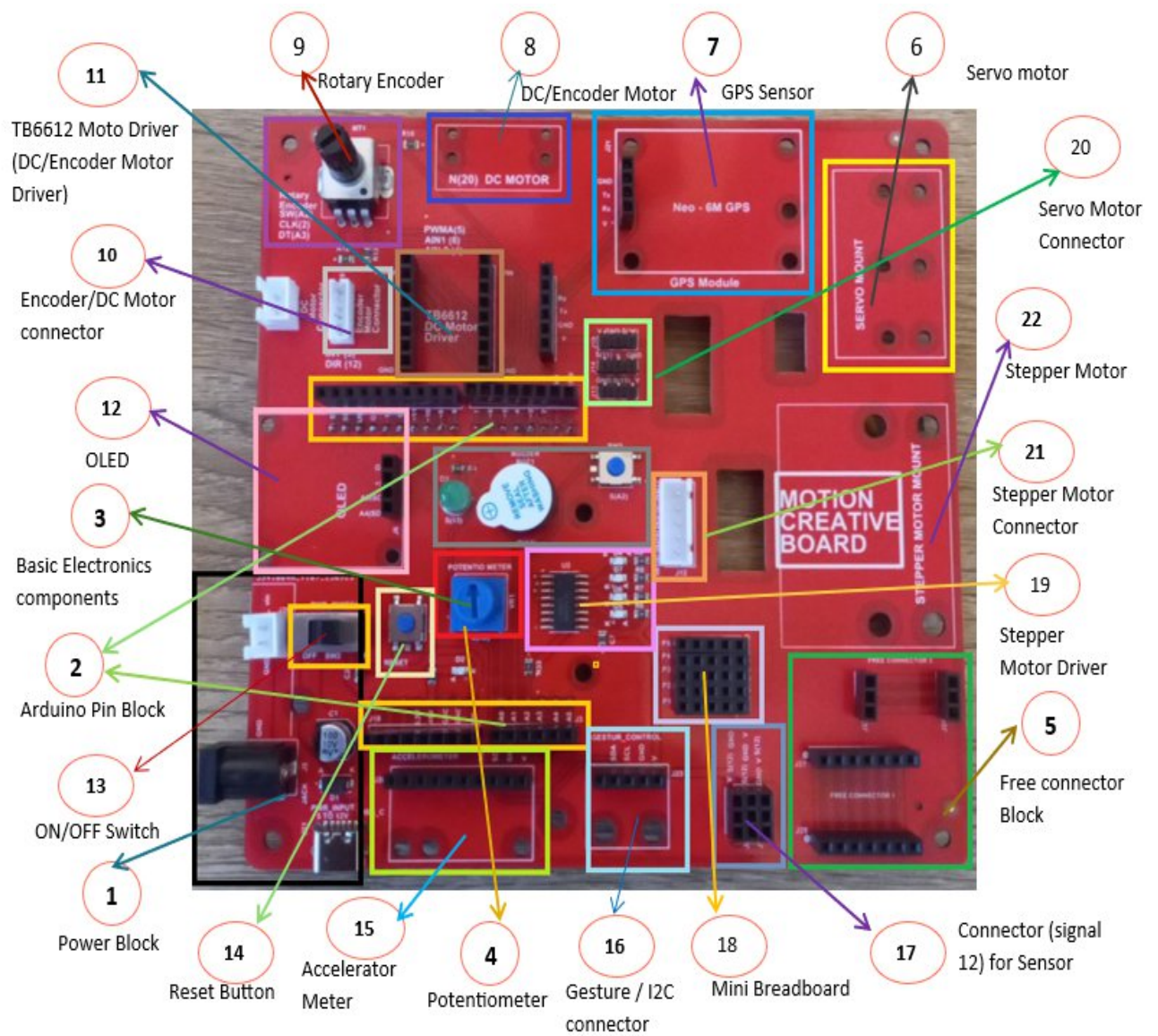


Figure 1. ZAS Robotics Motion Creative Board Functional Block Diagram

2. Power Supply Block

The Power Supply Block is responsible for providing stable and safe power to the entire ZAS Robotics Motion Creative Board.

Figure 2 highlights the power section, including all available input options and the onboard power switch.

The Motion Creative Board supports three different power input methods, allowing maximum flexibility for classroom and project use:

Supported Power Input Options

DC Jack Input (5V to 7.4V)

Suitable for external power adapters commonly used with Arduino-based projects.

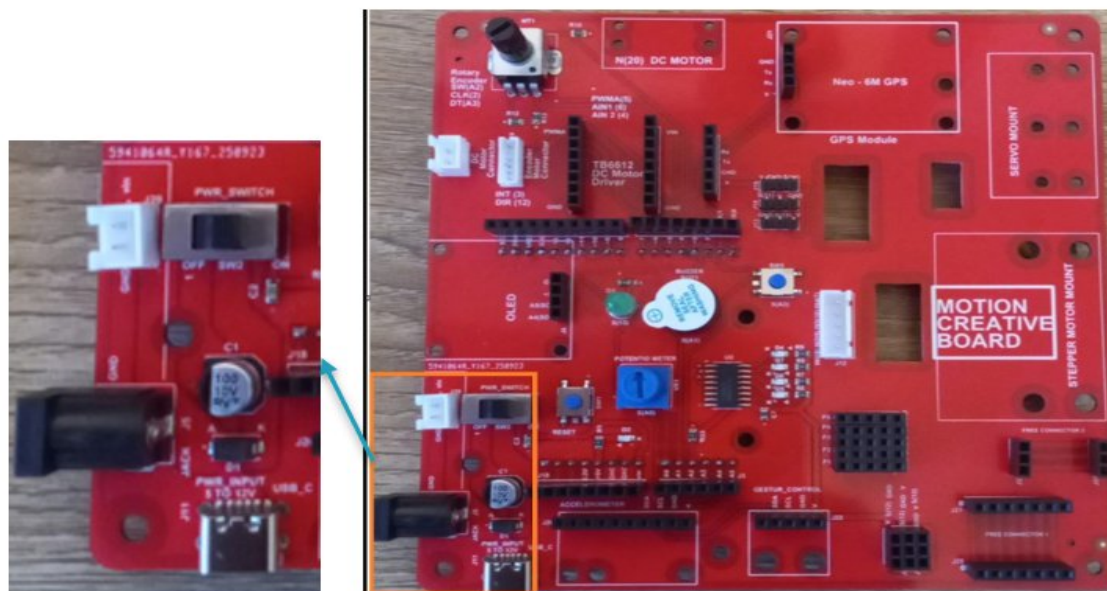
USB-C Cable (5V)

Allows powering the board from laptops, power banks, or USB chargers.

Ideal for programming, testing, and indoor sessions.

JST XH 2-Pin Battery Input (3.7V to 7.4V)

Supports Li-ion/Li-Po batteries for portable or mobile robotic applications.



Board can be powered from

- DC Jack or
- USB C Cable or
- JST XH 2 Pin Battery

Figure 2. ZAS Robotics Motion Creative Board – Power Supply Block

Power Control

The board includes a dedicated ON/OFF switch, which allows the user to completely disconnect the input power without unplugging cables.

This improves safety, prevents accidental short circuits, and helps in classroom demonstrations.

3. Arduino Block

Figure 3 shows the front view of the Motion Creative Board.

The long black pin headers highlighted in orange are Arduino Pin Exposure Blocks. These connectors do not contain electronics — they simply bring out (expose) the Arduino UNO pins to the front side of the board for easy access.

When an Arduino UNO is mounted at the back (Figure 4 and Figure 5), all of its I/O pins are internally routed to these front connectors. available Arduino Pins on the Front Panel

The following Arduino pins are fully exposed on the Motion Creative Board:

- Digital I/O Pins: D0 – D13
- Analog Input Pins: A0 – A5
- Power Pins: 5V, 3.3V, GND, VIN
- Communication Pins:
- I²C: SCL, SDA
- UART: RX, TX

This design ensures that sensors, modules, motors, and shields can be plugged directly on the front side without touching or removing the Arduino UNO from the back side of the board.

Reset Functionality

The board also includes a dedicated Reset push-button on the front panel. This button is electrically connected to the RESET pin of the Arduino UNO, allowing the user to restart the Arduino safely and conveniently from the front side — especially useful during programming and debugging.

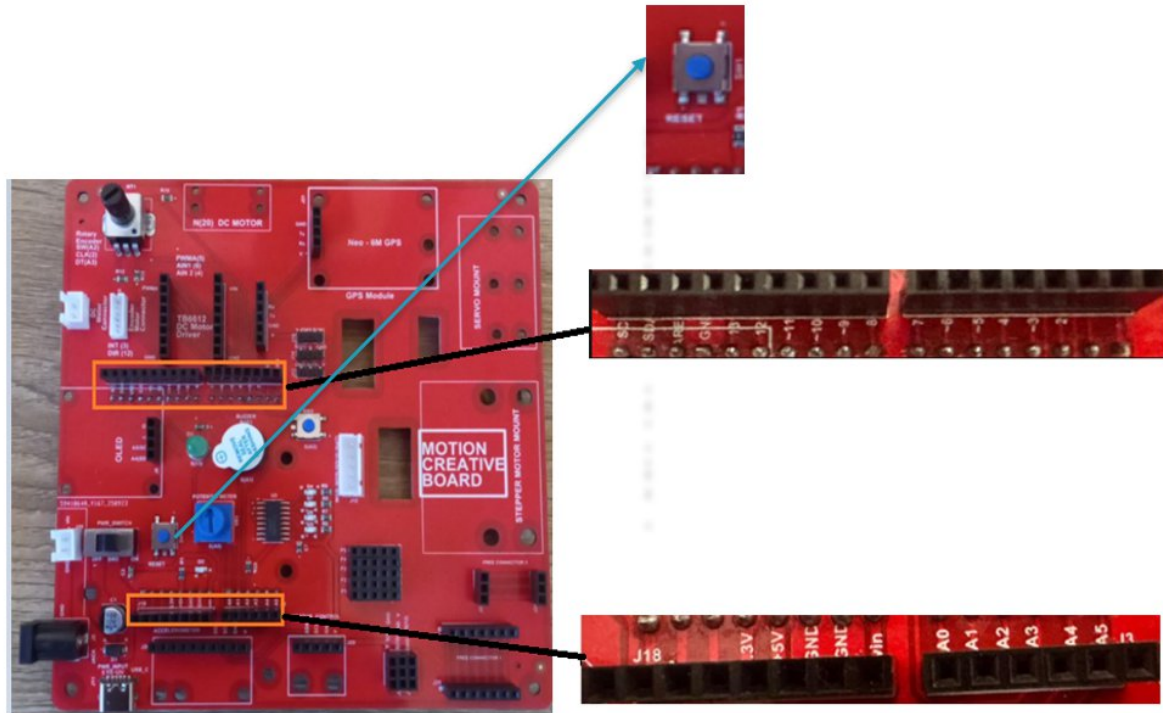


Figure 3. ZAS Robotics Motion Creative Board – Arduino Pins Block

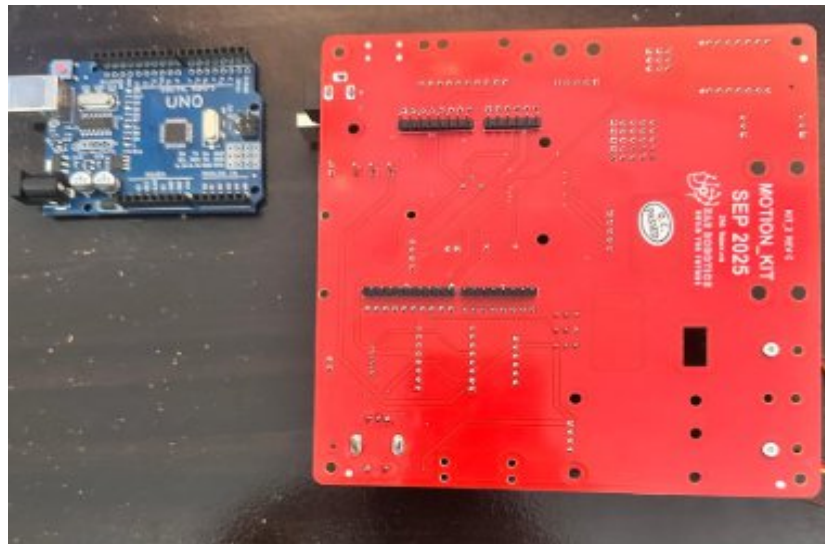


Figure 4. ZAS Robotics Motion Creative Board – Arduino unmounted

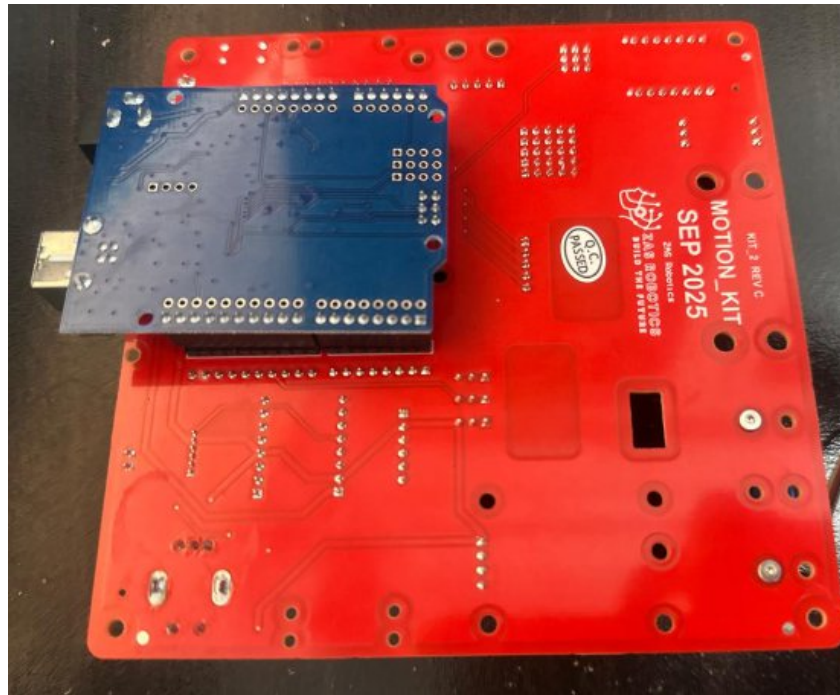


Figure 5. ZAS Robotics Motion Creative Board – Arduino Mounted

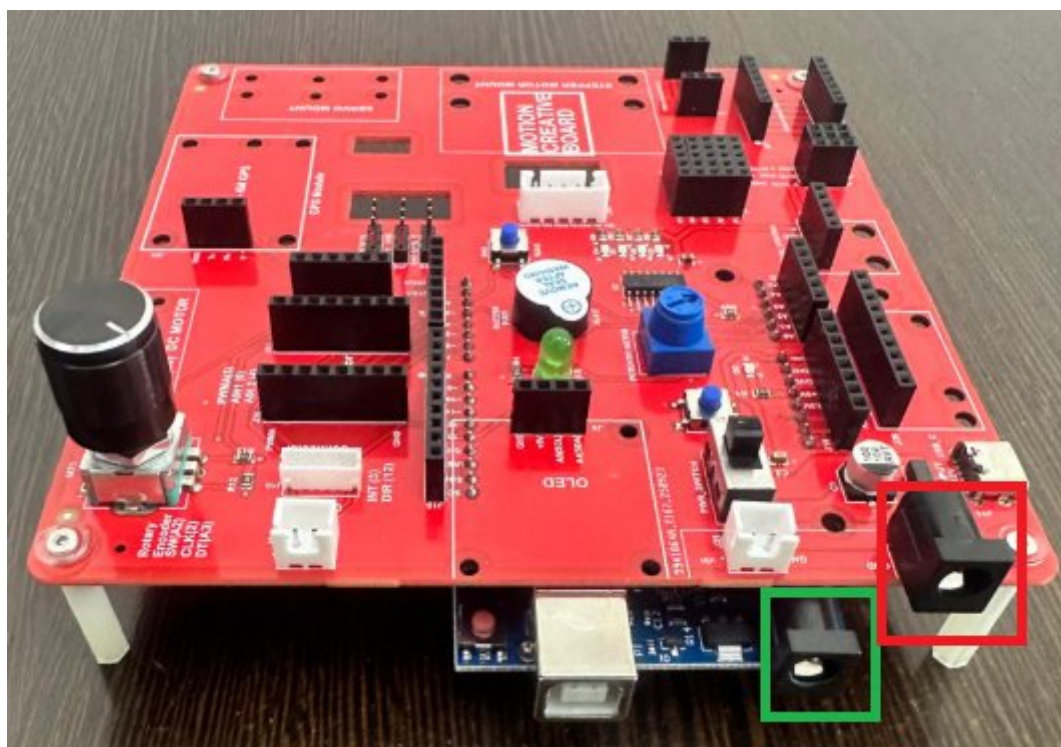


Figure 6. ZAS Robotics Motion Creative Board – Arduino Mounted

Important Note – Arduino Mounting Orientation

When mounting the Arduino UNO onto the Motion Creative Board, ensure that the Arduino's DC Jack aligns on the same side as the Motion Board's DC Jack, as shown in Figure 5. This alignment guarantees correct pin matching and safe electrical connection.

4. Basic Electronics Components

This area contains commonly used electronic components such as LEDs, test points, and simple I/O breakout pins. It allows students to learn basic digital and analog input/output experiments, LED blinking, resistor understanding, and fundamental electronics.



Figure 7. ZAS Robotics Motion Creative Board –Basic Electronics Components

Component	Arduino pin mapping
Active Buzzer	A1
Green LED	13
Reset Button	A2

5. Potentiometer

The onboard potentiometer provides an adjustable analog input to the Arduino. It is commonly used to control analog values, such as motor speed, brightness, or menu navigation.



Figure 8. ZAS Robotics Motion Creative Board –Potentiometer

Component	Arduino pin mapping
Potentiometer	A0

6. Free Connectors Block

General-purpose male/female connectors that allow to connect custom sensors, jumper wires, and prototypes. These connectors enable flexible experimentation and expansion beyond the predefined modules.

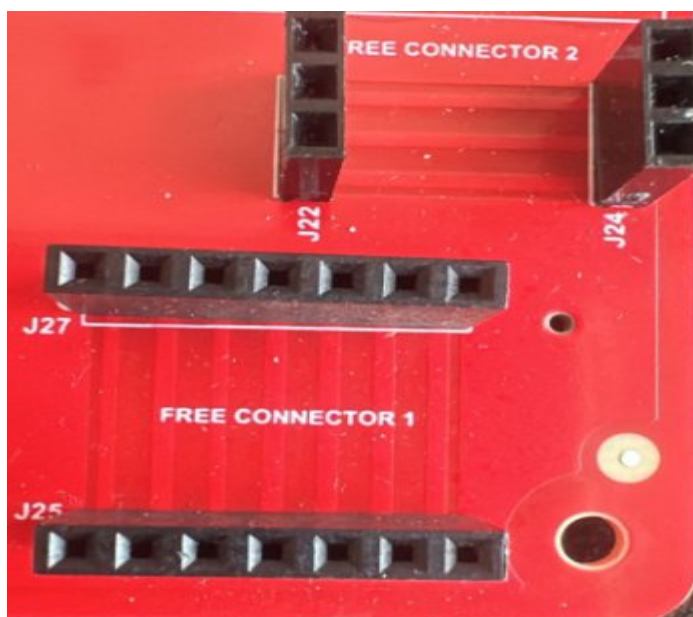


Figure 9. ZAS Robotics Motion Creative Board –Free Connectors Block

7. Servo Motor

A designated area for mounting a servo motor for applications like angle control, steering, and robotic arms.

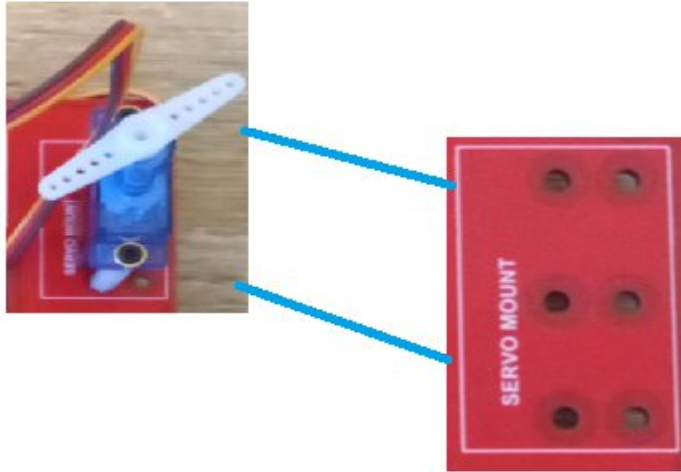


Figure 10. ZAS Robotics Motion Creative Board –Servo Motor

Component	Arduino pin mapping
Servo Motor	11

8. GPS Sensor

A dedicated slot for the NEO-6M GPS module, allowing students to explore GPS data such as latitude, longitude, and satellite tracking.

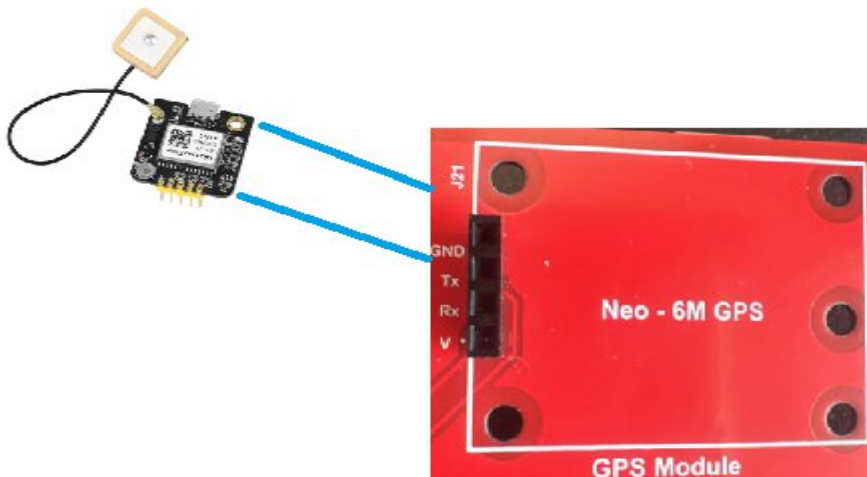


Figure 11. ZAS Robotics Motion Creative Board –GPS Sensor

Component	Arduino pin mapping
Tx	Rx

Rx	Tx
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9. DC/Encoder Motor

This area supports N20/N60 DC motors with encoders. It allows learning about direction control, PWM speed regulation, and reading encoder feedback for speed/position measurement.

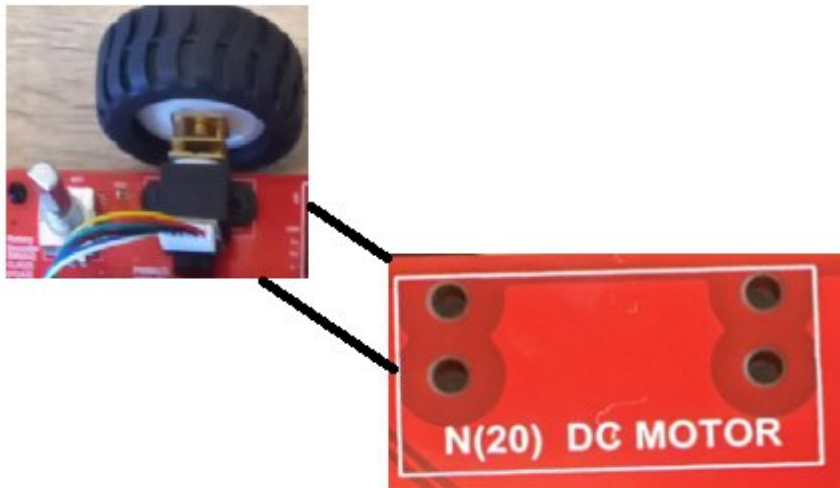


Figure 12. ZAS Robotics Motion Creative Board –DC/Encoder Motor

10. Rotary Encoder

A dial-type input device used to detect rotation. Useful for speed measurement, menu control, and hands-on learning of encoder signals (CLK, DT, SW).



Figure 13. ZAS Robotics Motion Creative Board –Rotary Encoder

Component	Arduino pin mapping
SW	A2
CLK	2

DT	A3
----	----

11. Encoder/DC Motor Connector

A 4-pin/6-pin connector that brings encoder signals and motor power lines from the motor to the TB6612 driver. Ensures clean and error-proof wiring.

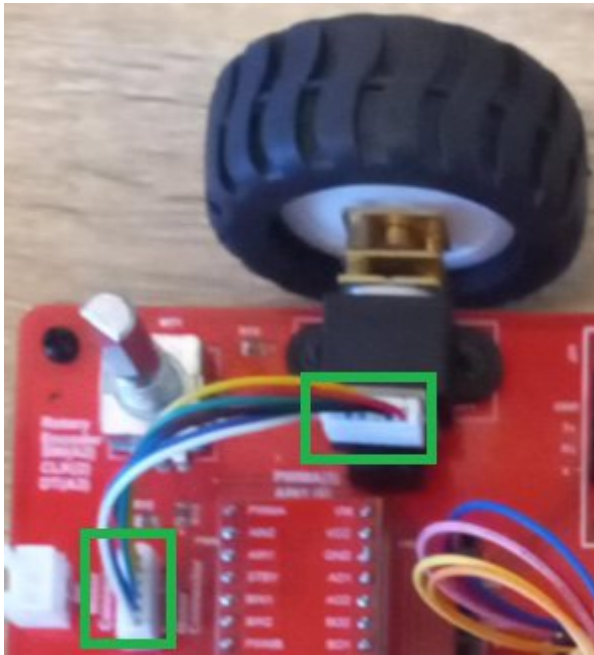


Figure 14. ZAS Robotics Motion Creative Board –Encoder/DC Motor Connector

Component	Arduino pin mapping
PWMA	5
IN1	6
IN2	4
DIR	12
ENCA	3

12. TB6612 Motor Driver (DC/Encoder Motor Driver)

A dual H-Bridge DC motor driver that controls motor direction and speed using PWM. Supports encoder-based motors and is widely used for robotics motion control.

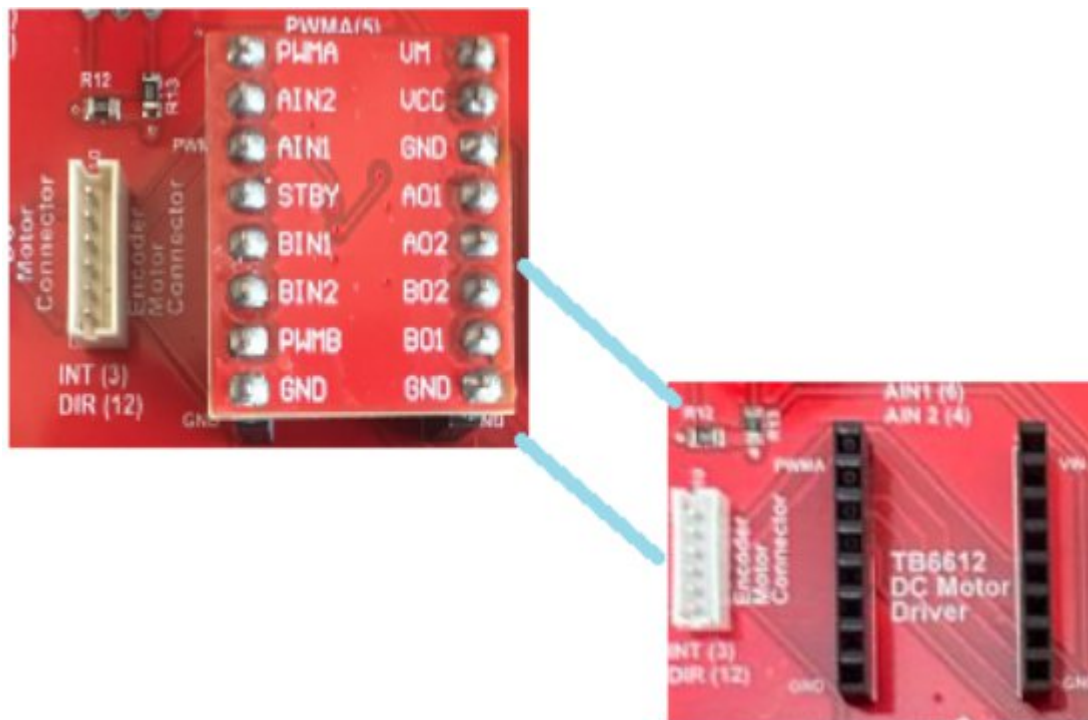


Figure 15. ZAS Robotics Motion Creative Board – TB6612 Motor Driver (DC/Encoder Motor Driver)

13. OLED

A 0.96" I²C OLED screen slot used for displaying text, sensor values, and debugging information.

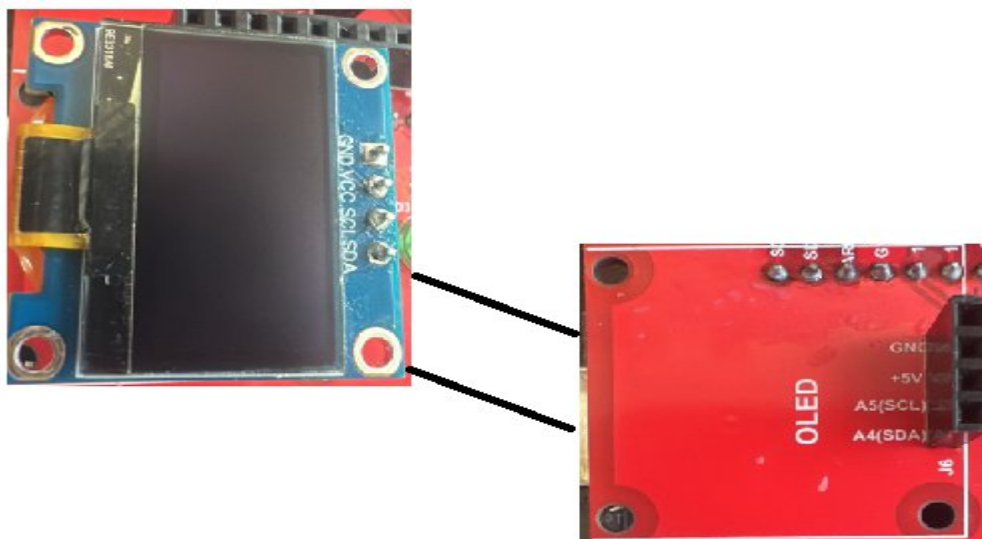


Figure 16. ZAS Robotics Motion Creative Board – OLED

Vcc	5V
GND	GND
SCL	A5
SDA	A4

14.ON/OFF Switch

Controls the main power input to the board. Ensures safety by allowing the user to disconnect power without unplugging cables.

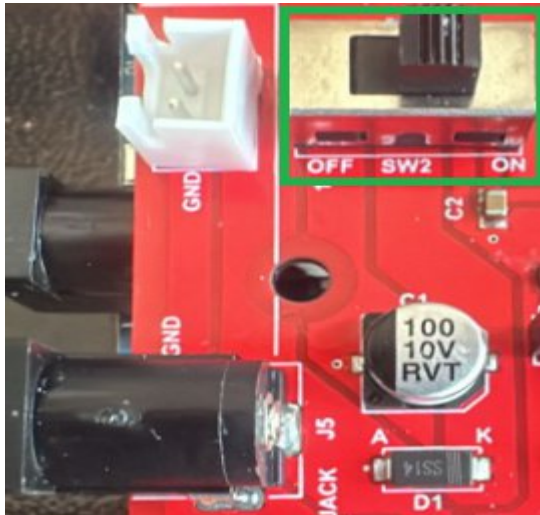


Figure 17. ZAS Robotics Motion Creative Board –ON/OFF Switch

15.Reset Button

A front-mounted button connected to the Arduino UNO RESET pin. Allows restarting the Arduino without removing or touching it at the back side.



Figure 18. ZAS Robotics Motion Creative Board –Reset Button

Component	Arduino pin mapping
Reset Button	A2

16.Accelerator

An analog accelerator / motion meter used for speed control or analog input experiments. It simulates throttle-like behavior.

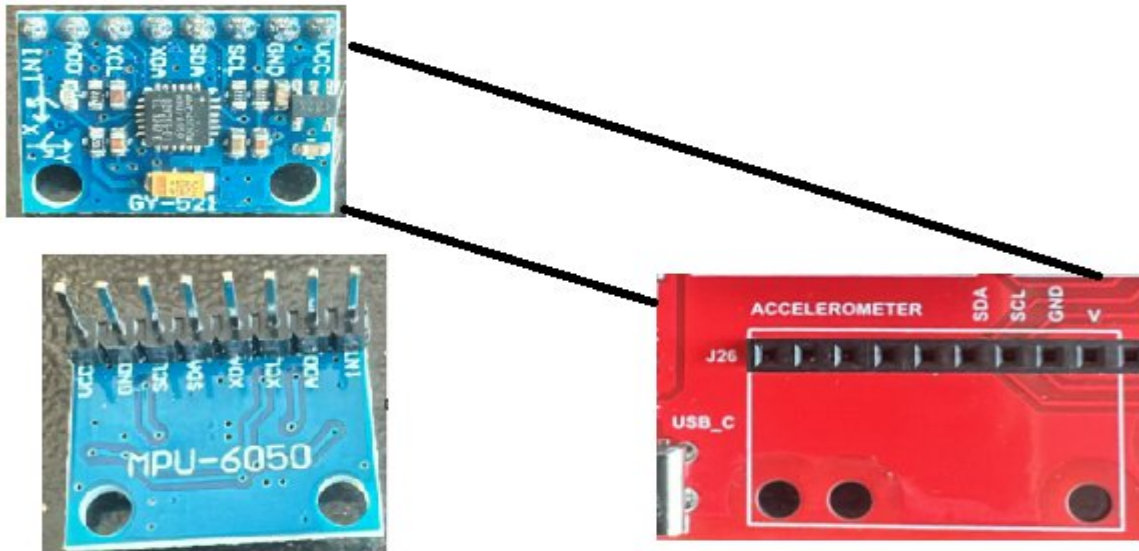


Figure 19. ZAS Robotics Motion Creative Board – Accelerator

17.Gesture / I2C Connector

A universal I²C header that supports gesture sensors, accelerometers, magnetometers, light sensors, and any 3.3V/5V I2C-compatible module.



Figure 20. ZAS Robotics Motion Creative Board – Gesture / I²C Connector

18. Sensor Connector (Signal Pin 12)

A special sensor connector wired to Digital Pin 12, designed for simple plug-in sensors such as IR, ultrasonic (via trigger/echo breakout), and digital switches.



Figure 21. ZAS Robotics Motion Creative Board – Connector (Signal Pin 12) for Sensor

Component	Arduino pin mapping
Sensor (Any)	12

19. Mini Breadboard

A small prototyping area for custom circuits, resistors, capacitors, homemade sensors, and experimental wiring.

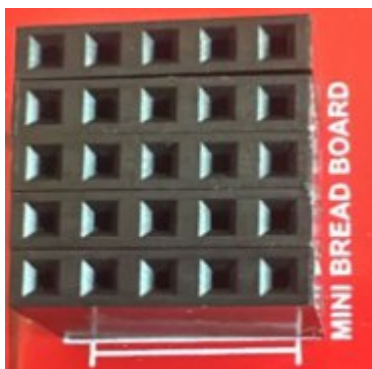


Figure 22. ZAS Robotics Motion Creative Board – Mini Breadboard

20. Stepper Motor Driver

A driver slot compatible with ULN2003 or A4988 modules, allowing precise control of stepper motors for robotics applications.

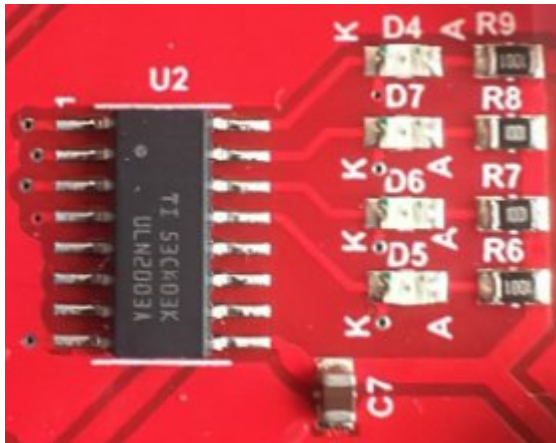


Figure 23. ZAS Robotics Motion Creative Board – Stepper Motor Driver

21. Servo Connector

A 3-pin header (5V–GND–Signal) for directly connecting servo motors. Supports standard servo control through Arduino PWM pins.



Figure 24. ZAS Robotics Motion Creative Board –Servo Connector

Component	Arduino pin mapping
Servo Connector	11

22. Stepper Motor Connector

A dedicated connector for 28BYJ-48 stepper motors or other compatible models. Ensures easy plug-and-play connection.

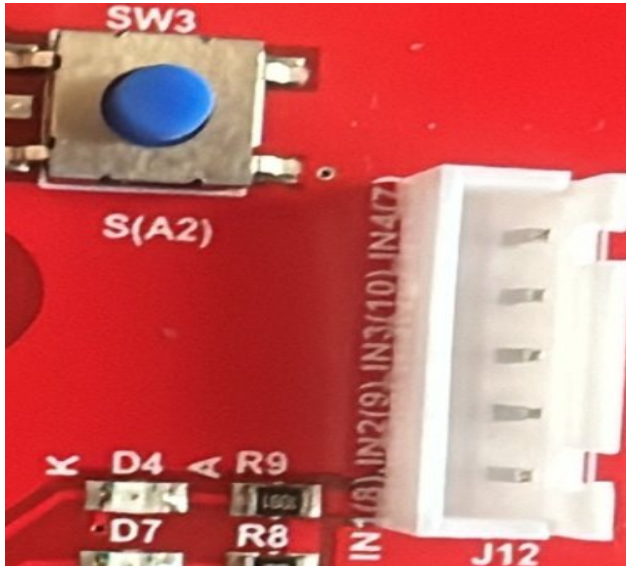


Figure 25. ZAS Robotics Motion Creative Board –Stepper Motor Connector

Component	Arduino pin mapping
IN1	8
IN2	9
IN3	10
IN4	7

23. Stepper Motor

The motor used for precise angular movement, rotation experiments, robotic joints, or position-controlled tasks.

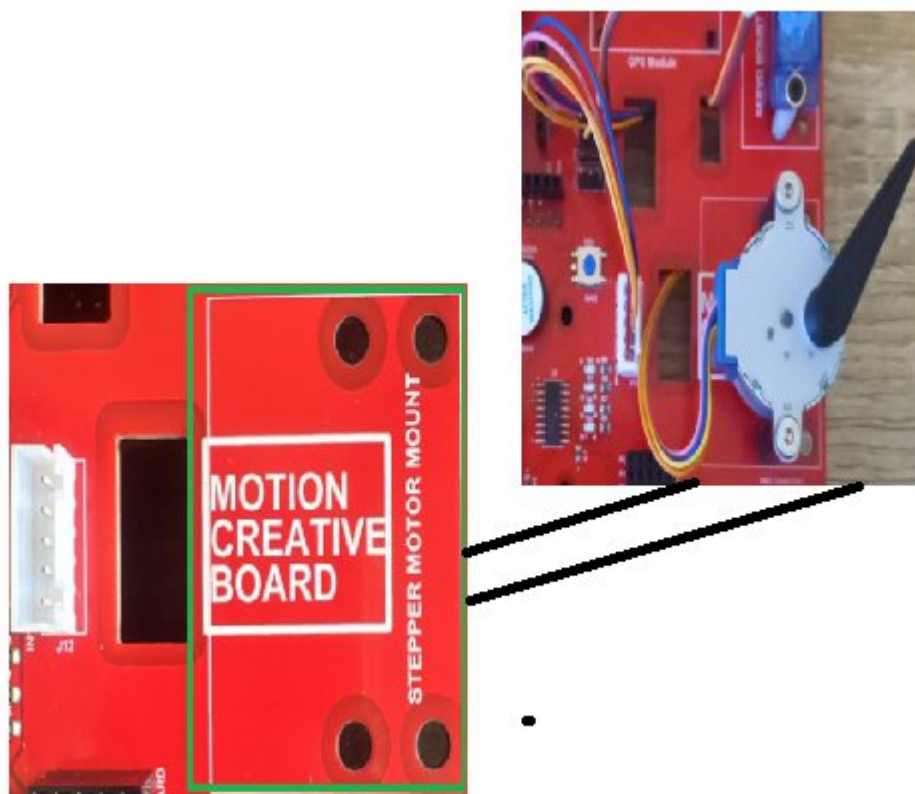


Figure 26. ZAS Robotics Motion Creative Board – Stepper Motor

Component	Arduino pin mapping
IN1	8
IN2	9
IN3	10
IN4	7